

MAXIMUM RATINGS

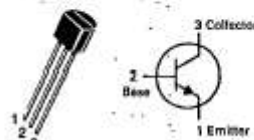
Rating	Symbol	Value	Unit
Collector-Emitter Voltage	V_{CEO}	- 40	Vdc
Collector-Base Voltage	V_{CBO}	60	Vdc
Emitter-Base Voltage	V_{EBO}	6.0	Vdc
Collector Current — Continuous	I_C	100	mAdc
Total Device Dissipation @ $T_A = 25^\circ\text{C}$ Derate above 25°C	P_D	625 5.0	mW mW/°C
Total Power Dissipation @ $T_A = 60^\circ\text{C}$	P_D	450	mW
Total Device Dissipation @ $T_C = 25^\circ\text{C}$ Derate above 25°C	P_D	1.5 12	Watts mW/°C
Operating and Storage Junction Temperature Range	T_J, T_{stg}	-55 to +150	°C

THERMAL CHARACTERISTICS

Characteristic	Symbol	Max	Unit
Thermal Resistance, Junction to Case	$R_{\theta JC}$	83.3	°C/W
Thermal Resistance, Junction to Ambient	$R_{\theta JA}$	200	°C/W

MPS3903
MPS3904

CASE 29-04, STYLE 1
TO-92 (TO-226AA)



**GENERAL PURPOSE
TRANSISTORS**

NPN SILICON

ELECTRICAL CHARACTERISTICS ($T_A = 25^\circ\text{C}$ unless otherwise noted.)

Characteristic	Symbol	Min	Max	Unit
OFF CHARACTERISTICS				
Collector-Emitter Breakdown Voltage(1) ($I_C = 1.0 \text{ mAdc}, I_B = 0$)	$V_{(BR)CEO}$	40	—	Vdc
Collector-Base Breakdown Voltage ($I_C = 10 \mu\text{A}, I_E = 0$)	$V_{(BR)CBO}$	60	—	Vdc
Emitter-Base Breakdown Voltage ($I_E = 10 \mu\text{A}, I_C = 0$)	$V_{(BR)EBO}$	6.0	—	Vdc
Collector Cutoff Current ($V_{CE} = 30 \text{ Vdc}, V_{EB(off)} = 3.0 \text{ Vdc}$)	I_{CEX}	—	50	nA
Base Cutoff Current ($V_{CE} = 30 \text{ Vdc}, V_{EB(off)} = 3.0 \text{ Vdc}$)	I_{BL}	—	50	nA
ON CHARACTERISTICS(1)				
DC Current Gain ($I_C = 0.1 \text{ mA}, V_{CE} = 1.0 \text{ Vdc}$)	h_{FE}	MPS3903 MPS3904	20	—
			40	—
($I_C = 1.0 \text{ mA}, V_{CE} = 1.0 \text{ Vdc}$)		MPS3903 MPS3904	35	—
			70	—
($I_C = 10 \text{ mA}, V_{CE} = 1.0 \text{ Vdc}$)		MPS3903 MPS3904	50	150
			100	300
($I_C = 50 \text{ mA}, V_{CE} = 1.0 \text{ Vdc}$)		MPS3903 MPS3904	30	—
			60	—
($I_C = 100 \text{ mA}, V_{CE} = 1.0 \text{ Vdc}$)		MPS3903 MPS3904	15	—
			30	—
Collector-Emitter Saturation Voltage ($I_C = 10 \text{ mA}, I_B = 1.0 \text{ mA}$) ($I_C = 50 \text{ mA}, I_B = 5.0 \text{ mA}$)	$V_{CE(sat)}$	—	0.2 0.3	Vdc
Base-Emitter Saturation Voltage ($I_C = 10 \text{ mA}, I_B = 1.0 \text{ mA}$) ($I_C = 50 \text{ mA}, I_B = 5.0 \text{ mA}$)	$V_{BE(sat)}$	0.65 —	0.85 1.1	Vdc

MOTOROLA SC XSTRS/R F

12E D

6367254 0086082 3

MPS3903, MPS3904

T-27-15

ELECTRICAL CHARACTERISTICS (continued) ($T_A = 25^\circ\text{C}$ unless otherwise noted.)

Characteristic	Symbol	Min	Max	Unit
SMALL-SIGNAL CHARACTERISTICS				
Current-Gain — Bandwidth Product ($I_C = 10\text{ mAdc}$, $V_{CE} = 20\text{ Vdc}$, $f = 100\text{ kHz}$)	f_T	150 200	—	MHz
Output Capacitance ($V_{CB} = 5.0\text{ Vdc}$, $I_E = 0$, $f = 100\text{ kHz}$)	C_{obo}	—	4.0	pF
Input Capacitance ($V_{BE} = 0.5\text{ Vdc}$, $I_C = 0$, $f = 100\text{ kHz}$)	C_{ibo}	—	8.0	pF
Input Impedance ($I_C = 1.0\text{ mAdc}$, $V_{CE} = 10\text{ Vdc}$, $f = 1.0\text{ kHz}$)	h_{ie}	0.5 1.0	8.0 10	k Ω
Voltage Feedback Ratio ($I_C = 1.0\text{ mAdc}$, $V_{CE} = 10\text{ Vdc}$, $f = 1.0\text{ kHz}$)	h_{ve}	0.1 0.5	5.0 8.0	$\times 10^{-4}$
Small-Signal Current Gain ($I_C = 1.0\text{ mAdc}$, $V_{CE} = 10\text{ Vdc}$, $f = 1.0\text{ kHz}$)	h_{fa}	50 100	200 400	—
Output Admittance ($I_C = 1.0\text{ mAdc}$, $V_{CE} = 10\text{ Vdc}$, $f = 1.0\text{ kHz}$)	h_{oe}	1.0	40	μmhos
Noise Figure ($I_C = 100\text{ }\mu\text{Adc}$, $V_{CE} = 5.0\text{ Vdc}$, $R_S = 1.0\text{ k}\Omega$, $f = 10\text{ Hz to }15.7\text{ kHz}$)	NF	—	6.0 5.0	dB
SWITCHING CHARACTERISTICS				
Delay Time	t_d	—	35	ns
Rise Time	t_r	—	50	ns
Storage Time	t_s	—	800 900	ns
Fall Time	t_f	—	90	ns

(1) Pulse Test: Pulse Width $\leq 300\text{ }\mu\text{s}$, Duty Cycle $\leq 2.0\%$.

EQUIVALENT SWITCHING TIME TEST CIRCUITS

FIGURE 1 — TURN-ON TIME

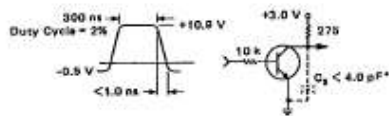
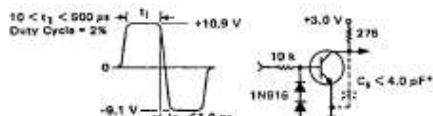


FIGURE 2 — TURN-OFF TIME



*Total shunt capacitance of test jig and connectors

FIGURE 8 – DC CURRENT GAIN

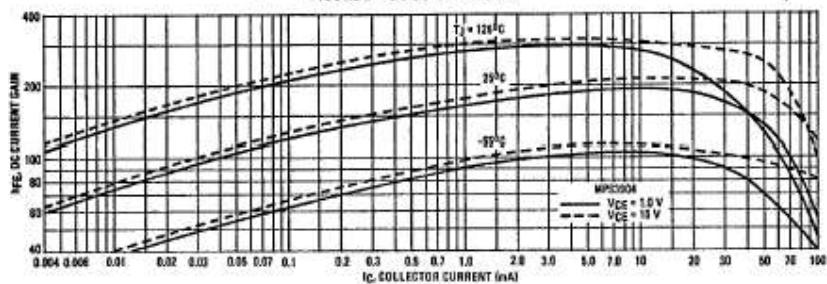


FIGURE 9 – COLLECTOR SATURATION REGION

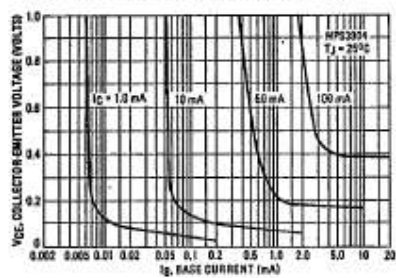


FIGURE 10 – COLLECTOR CHARACTERISTICS

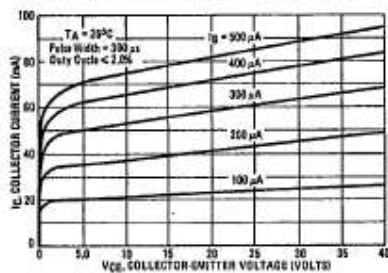


FIGURE 11 – "ON" VOLTAGES

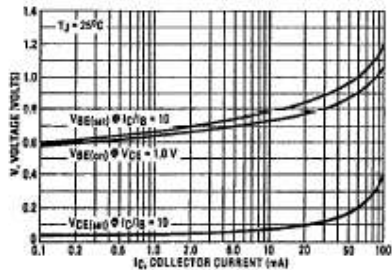


FIGURE 12 – TEMPERATURE COEFFICIENTS

